Digital Electronics (ECT-033) - Unit 5

Unit 5 - VLSI Design Flow & VHDL - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: how VLSI chips are designed, and VHDL - the language used to describe digital hardware.

Key ideas: The **VLSI design flow**: specification → design entry (VHDL/schematic) → functional simulation → synthesis → place and route → timing verification → fabrication. Know the difference between simulation levels (behavioural, RTL, gate). **VHDL** basics: every design has an **entity** (ports: in, out, inout) and an **architecture** (the internal logic). Modelling styles: **dataflow** (concurrent signal assignments), **behavioural** (process with if/case - sequential statements), **structural** (connecting components like a netlist). Learn the skeleton of one example each: a 2:1 MUX, a full adder, a D flip-flop.

Exam pointers: "write VHDL code for a full adder / 2:1 MUX / D flip-flop, any modelling style" is the standard coding question - memorise the entity-architecture skeleton once and reuse it. Theory: design-flow diagram and the three modelling styles with one-line differences.

Made by Anas. Free forever. Watch the linked videos on the site after reading.